

BENCHMARK CONSTRUCTION DETAILS

This appendix specifies the twenty-task composite disaster-response benchmark used in Section V-A of the main paper. Each task is a deterministic function of a single integer index. The auxiliary data files needed to reproduce the test system are released alongside the paper¹. The four dimensions (d, δ, τ, ρ) denote travel-demand intensity, transportation-network damage, distribution-network topology faults, and renewable-generation availability. Each dimension is varied independently, so tasks remain comparable along each axis.

A. Task Construction

1) *Task Index and the Four-Dimensional Encoding*: Each task is identified by a mode index $m \in \mathcal{M} \triangleq \{0, \dots, 19\}$. The three deterministic dimensions are obtained from m through

$$\begin{aligned} d(m) &= \lfloor m/4 \rfloor + 1, \\ \tau(m) &= m \bmod 4, \\ \rho(m) &= m \bmod 10, \end{aligned} \quad (1)$$

so that $d(m) \in \{1, \dots, 5\}$, $\tau(m) \in \{0, \dots, 3\}$, and $\rho(m) \in \{0, \dots, 9\}$. The transportation-network disruption $\delta(m)$ is generated independently for each m as defined in Section -B1. The encoding sweeps the OD demand index d over its five levels and the PDN topology index τ over its four levels. This yields twenty distinct (d, τ) combinations. The diurnal PV index ρ cycles twice across $\mathcal{M}$, but no two tasks share the same (d, τ, ρ) triple. Tasks $m \in \{0, \dots, 14\}$ form the in-distribution (IID) training split $\mathcal{M}_{\text{IID}}$. Tasks $m \in \{15, \dots, 19\}$ form the out-of-distribution (OOD) held-out split $\mathcal{M}_{\text{OOD}}$, comprising the configuration $(d, \tau) = (4, 3)$ at $m = 15$ and the four $d = 5$ configurations at $m \in \{16, \dots, 19\}$.

2) *OD Demand Levels*: The five OD demand levels $d_1, \dots, d_5$ are stored as integer traveler counts per OD pair. The value $d(m)$ in (1) selects one level at runtime. The eight OD pairs connect the origin set $\{1, 2, 3, 7\}$ to the destination set $\{6, 10, 11, 12\}$ and are common to all tasks. Mean demand across the eight OD pairs varies between 28.4 and 88.6 travelers across the five levels. This gives a roughly threefold range on a common network topology. A diurnal multiplier modulates the base demand. It peaks near 2.28 in the morning rush, decreases to 0.57–0.76 around midday, and rises to about 2.05 in the evening rush. The effective travel demand entering the traffic-assignment layer is the element-wise product of the selected level and this multiplier.

B. Disaster Scenarios

1) *Transportation-Network Disruption*: The transportation-network disruption $\delta(m)$ is represented as a time-indexed capacity-reduction field $\sigma_m : \mathcal{L} \times \mathcal{T} \rightarrow (0, 1]$. Here, $\mathcal{L}$ denotes the set of $|\mathcal{L}| = 40$ directed links, and $\mathcal{T} = \{0, \dots, 11\}$ denotes the restoration horizon. For every $m \in \mathcal{M}$, an independent realization is generated according to

$$\sigma_m(\ell, t) = \begin{cases} U_{m,\ell}, & (\ell, t) \in \mathcal{L}_m \times \{t_m\}, \\ 1, & \text{otherwise,} \end{cases} \quad (2)$$

where t_m is drawn uniformly from the disruption window $\{3, \dots, 10\}$. The damaged link subset $\mathcal{L}_m \subseteq \mathcal{L}$ is obtained by sampling a damage ratio $\pi_m \sim \mathcal{U}(0.2, 0.4)$ and selecting $\lceil \pi_m |\mathcal{L}| \rceil$ links uniformly without replacement. The residual-capacity multipliers $U_{m,\ell} \sim \mathcal{U}(0.2, 0.4)$ are drawn independently across $\ell \in \mathcal{L}_m$. Each task contains a single localized shock. A fraction of the network is degraded for one hour and restored thereafter. This one-hour shock keeps the disruption model simple and is consistent with single-point hazard abstractions used in short-horizon distribution restoration studies. Sustained or cascading disruptions are outside the scope of this benchmark. The sampling procedure uses a fixed random seed, so the resulting disruption field is identical across training and evaluation runs.

2) *Distribution-Network Topology Faults*: The four PDN topology configurations $\tau_0, \dots, \tau_3$ correspond to distinct $N-k$ branch-outage scenarios on the modified IEEE 33-bus feeder. Each configuration is stored as a full branch-parameter table. At solve time, the affected branches are removed from the power-flow feasibility set by inflating their impedances above a large threshold. Configurations τ_0, τ_1 , and τ_3 are $N-1$ contingencies. They open one branch in the upstream lateral, a mid-feeder lateral, and a branch adjacent to a heavy-load node, respectively. Configuration τ_2 is an $N-3$ contingency that simultaneously opens the primary tie and two laterals. The four configurations form a graded sequence in the number of unserved downstream load buses. They provide a controlled ordering from mild to severe PDN fault regimes while keeping the restored network radial in every case.

3) *Photovoltaic Profiles*: The ten PV profiles $\rho_0, \dots, \rho_9$ are daily normalized power curves obtained from a fifteen-minute irradiance record. They are downsampled to hourly resolution across the restoration horizon and applied uniformly to the five PV buses of the feeder. The profiles span conditions from high-yield clear-sky days with a sharp solar-noon peak to heavily intermittent days with rapid mid-day drop-outs. Thus, renewable availability forms a distinct component of the task context. At dispatch time, each curve is scaled by the installed PV capacity, aggregating to 3.8 MW across the five PV buses. Because $\rho(m) = m \bmod 10$, each profile is exercised exactly twice across the benchmark. Each recurrence is paired with a different (d, τ) combination, preserving task distinctness.

C. Test Systems

1) *Distribution Feeder*: The distribution side is a modified IEEE 33-bus feeder with thirty-two branches, voltage bounds $[0.9, 1.1]$ p.u., and a slack bus at node 0. Non-EV loads are grouped into four priority classes. The relative value-of-lost-load (VoLL) weights are $(2, 3, 4, 5)$. Time-invariant portions $(0.87, 0.71, 0.84, 0.77)$ of the hourly nodal demand are assigned to the industrial, commercial, residential, and critical classes, respectively. EV charging and V2G injections enter as nodal exchanges with a penalty weight equal to twice the prevailing electricity price. Within the restoration window, the energy price used by the optimal-power-flow objective is held at 0.38 USD/kWh. This removes intra-horizon price variation and isolates the effect of the four varied dimensions.

¹https://github.com/eityh41-oss/Meta_RL_Extreme_events

2) *Transportation Network and Charging Layer*: The transportation side is a twelve-node network with forty directed links and eight commuting OD pairs. Ten charging stations co-locate with selected transportation nodes. There are four slow chargers at nodes $\{1, 2, 3, 7\}$ and nine fast chargers at $\{2, 3, 4, 5, 6, 7, 8, 9, 11\}$. Three nodes host both a slow and a fast charger. The two layers couple through a fixed charger-to-bus map. As a result, the ETAP-UE equilibrium determines the time-varying nodal injections seen by the PDN. Each EV cohort enters the horizon with an OD-pair-specific initial SoC in $[0.30, 0.45]$, averaging 0.33 across the eight pairs. This models a post-disaster vehicle population that has not recently had the opportunity to recharge.

D. Offline Data and Reproducibility

1) *Offline Corpus and OOD Context Sampling*: The offline corpus contains one behavior-policy trajectory collection per task. For each $m \in \mathcal{M}$, a Soft Actor-Critic behavior policy is trained online against the corresponding environment, and its rollouts are stored. The collection seed is fixed. Meta-training uses only the IID subset $\mathcal{M}_{\text{IID}}$. Three independent meta-training seeds are layered on top of this shared corpus. The encoder receives a single trajectory per IID task drawn from that task’s own behavior policy.

OOD evaluation additionally introduces a behavior-policy distribution shift. Rather than replaying stored trajectories for the held-out tasks, the evaluation driver performs a fresh rollout in each OOD environment. The behavior policy is drawn from a candidate pool composed exclusively of IID-trained SAC checkpoints. Replaying stored trajectories would leak the target-task action distribution through the context. The adopted protocol instead requires the encoder to infer task context from action distributions it has not observed at meta-training time. This protocol is used to evaluate both task-level and behavior-policy-level generalization. The OOD return reported in Section V-B of the main paper is obtained under this protocol.

2) *Summary of Released Parameters*: Table I lists the fixed parameters of the test systems and the disruption-sampling distributions, together with their derived properties. Together with the encodings in (1) and (2), these parameters fully define the twenty tasks. The auxiliary data files are released alongside the paper¹.

TABLE I
FIXED PARAMETERS OF THE TWENTY-TASK BENCHMARK.

Parameter	Value
Restoration horizon $ \mathcal{T} $	12 (hourly steps)
Voltage limits	$[0.9, 1.1]$ p.u.
PV installed capacity (aggregate)	3.8 MW
VoLL weights (ind./com./res./crit.)	2 : 3 : 4 : 5
EV-cut penalty relative to spot price	$2\times$
Electricity price over the horizon	0.38 USD/kWh
Disruption hour window	$\{3, \dots, 10\}$
Damage ratio / residual-capacity range	$[0.2, 0.4]$
SAC data-collection seed	10
Meta-training seeds	$\{110, 210, 310\}$